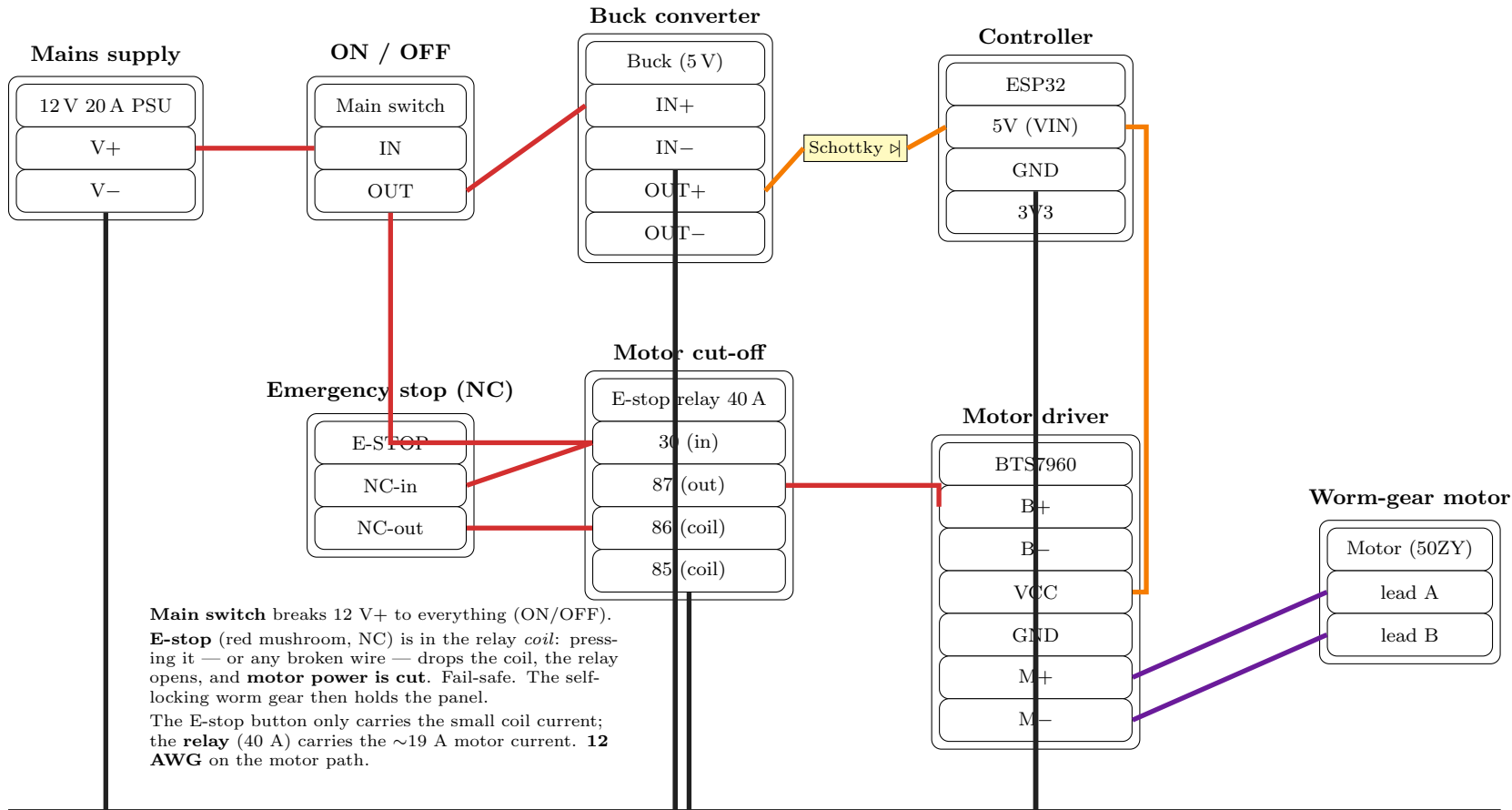


SolarMax — Wiring Diagrams

Colored wires by net. All grounds are common. Pin names match include/config.h.

Wire color legend: — +12 V — GND — +5 V — +3.3 V — Motor out — Motor PWM/EN — I²C — Limit switch — Anemometer sig — Analog (pot/LDR)

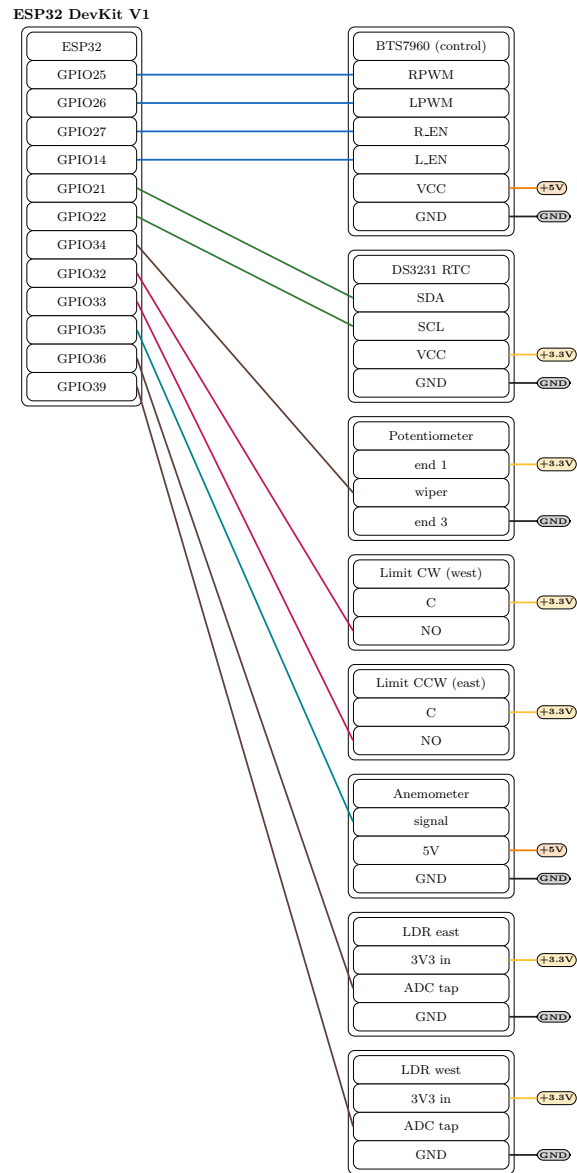
1. Power Distribution



ESP32 is powered from the buck via the **5V pin**. The **Schottky diode** lets you plug USB in any-time to flash. **USB = programming only**.

2. Control, Sensor & Power/Ground Wiring

Signal wires go left to the ESP32. Every power and ground pin is shown on the right with a colored tag. All GND tags are the same common ground.



Every GND tag is the one common ground. +5V and +3.3V come from the rails in Diagram 1. Limit switches: C → +3.3V, NO → GPIO (no GND pin). The LDR and anemometer taps are the centre of the resistor dividers in Diagram 3.

3. Voltage Divider Details

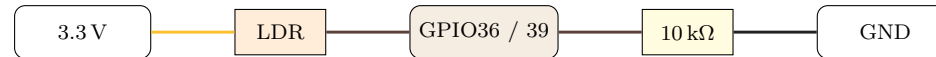
Two sensors need a resistor divider. Build these exactly — they protect the ESP32 and set the correct readings.

Anemometer signal: 5 V → 3.3 V



$$V_{out} = 5\text{ V} \times \frac{20}{10+20} \approx 3.3\text{ V}$$

LDR (each of the two): 3.3 V — LDR — ADC — 10 kΩ — GND



More light → lower LDR resistance
→ higher ADC reading